Mississippi Valley State University

CS 341 — Discrete Structures

Lecture 31: Counting Without Listing

The Multiplication and Addition Principles

Chartrand & Zhang, Section 8.1, pp. 271–280

Name: _____ Date: _____

You have been counting all term

The first thing you did in this course was count four-character binary passwords: 2^4. In Week 4 you counted subsets, 2^n, and Cartesian products, $|A| \cdot |B|$. In Week 3 you saw that SAT is hard because a formula has 2^n rows.
Today those arguments get names. Chapter 8 is less a new subject than the rules you have already been using, stated precisely — including when they fail.

By the end of today you should be able to

1. State and apply the Multiplication and Addition Principles.
2. Decide which one a problem calls for.
3. Recognise when the number of choices at one stage depends on an earlier stage.

1. The Multiplication Principle

The Multiplication Principle

A procedure consists of a sequence of tasks $T_1, T_2, \ldots, T_m$. If there are n_i ways to perform T_i **after** the preceding tasks have been performed, then the procedure can be performed in

$$n_1\, n_2 \cdots n_m \text{ ways.}$$

The signal word is "and then." First do this, then do that.

Count	Answer
Bit strings of length 8	$2^8 = 256$
Licence plates: 3 letters then 3 digits	$26^3 \cdot 10^3 = 17{,}576{,}000$
Four-digit numbers (first digit not 0)	$9 \cdot 10^3 = 9000$
Four-digit numbers with no repeated digit	$9 \cdot 9 \cdot 8 \cdot 7 = 4536$

Look at the last row

The second digit has 9 choices, not 10 — it may now be 0, but may not repeat the first. The count at each stage depends on what came before, and that is exactly why the principle says "after the preceding tasks have been performed."
The principle still works *as long as the number of choices is the same* no matter which earlier choices were made. When it is not, you need something more — see Section 3.

2. The Addition Principle

The Addition Principle

A procedure consists of performing **one** of the tasks $T_1, \ldots, T_m$, **no two of which can be performed at the same time**. If T_i can be performed in n_i ways, the procedure can be performed in $n_1 + n_2 + \cdots + n_m$ ways.

The signal word is "or" — with the options not overlapping.

Example. A lunch special offers 4 soups or 5 salads. Choices: 9.

Example. Strings over $\{a, b\}$ of length 1, 2, or 3: $2 + 4 + 8 = 14$.

Both are set facts you already have

Multiplication is $|A \times B| = |A| \cdot |B|$ — Lecture 11. A two-stage choice is an ordered pair.
Addition is $|A \cup B| = |A| + |B|$ when $A \cap B = \emptyset$ — disjointness from Lecture 12. "No two can be

performed at once" is disjointness.

3. When the stages interfere

How many three-digit even numbers have all distinct digits?

The tempting approach: choose the last digit (5 even options), then the others. But whether the last digit is 0 changes how many choices remain for the first digit, which cannot be 0. The count at stage one is **not fixed** — so split into cases.

Case	Count
Last digit is 0	first: 9 choices, middle: 8 $\Rightarrow 9 \cdot 8 = 72$
Last digit is $2, 4, 6$, or 8	last: 4, first: 8 (not 0, not last), middle: 8 $\Rightarrow 256$

The cases are disjoint, so by the Addition Principle the total is $72 + 256 = 328$.

Proof by cases, inside a counting problem

When the number of choices at some stage depends on an earlier choice, **split into cases that fix the troublesome choice**, count each with the Multiplication Principle, and add them with the Addition Principle.
That is Lecture 24's technique. Real counting problems use both principles together far more often than either alone.

Exit ticket

1. How many three-digit odd numbers are there? Say which principle you used.

2. Why can the three-digit even-number count not be done with a single product?

Before Wednesday

Read Section 8.2, "The Principle of Inclusion-Exclusion," pp. 280–286.

Self-check (not collected). Section 8.1, Exercises 1, 3, 5, and 7 — odd-numbered, answers in the back.

Coming up

Homework 11 is due Friday, November 6, 11:59 p.m.
Quiz 10 is Monday, November 9 — Sections 8.1 and 8.2.

Questions I still have